# TENDE: Transfer Entropy Neural Diffusion Estimation

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1. **Background**
   - Mutual Information (MI), Conditional Mutual Information (CMI)
   - Transfer Entropy (TE)

2. **Methods**
   - Score-based KL estimation
   - Score-based Entropy estimation
   - Score-based CMI and TE estimation

3. **Results**
   - Synthetic benchmark
   - Real data analysis

4. **Conclusion**

# Mutual Information (MI)

## Definition

Mutual Information (MI) between r.v. $X \in \mathbb{R}^{N_x}$ and $Y \in \mathbb{R}^{N_y}$:

$$I(X;Y) = D_{\mathrm{KL}}\left[p_{X,Y} \| p_X p_Y\right],$$

- $p_{X,Y}$: joint density ; $p_X$ and $p_Y$: marginal densities
- Kullback-Leibler (KL) Divergence between the distributions $p$ and $q$:

$$D_{\mathrm{KL}}[p\|q] = \mathbb{E}_{x\sim p}\left[\log\left(\frac{p(x)}{q(x)}\right)\right].$$

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## Remark

- $I(X;Y) = 0 \Leftrightarrow p_{X,Y} = p_X p_Y \Leftrightarrow X \perp Y$
- $I(X;Y) = I(Y;X)$
- MI only captures uncond. pairwise dependency

# Conditional Mutual Information (CMI)

## Definition

Conditional Mutual Information (CMI) quantifies the average MI between two r.v. $X$ and $Y$ given a third r.v. $Z$:

$$I(X; Y \mid Z) = \mathbb{E}_{z \sim p_Z}\left[I(X_z; Y_z)\right]$$

- $X_z \triangleq X \mid Z = z, \quad Y_z \triangleq Y \mid Z = z$

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## Remark

- For each fixed $z$,

$$I(X_z;Y_z) = D_{\mathrm{KL}}[p_{X_z,Y_z} \,\|\, p_{X_z}p_{Y_z}]\,,$$

  so $I(X;Y \mid Z)$ is the average KL divergence over $z \sim p_Z$.
- $I(X;Y \mid Z) = I(Y;X \mid Z) \Rightarrow$ does not capture directionality

# Transfer Entropy (TE)

## Definition

Let $\{X_t\}$ and $\{Y_t\}$ be two time series, and define

$$\mathbf{Y}_{t-\ell} = [Y_{t-1}, \ldots, Y_{t-\ell}], \qquad \mathbf{X}_{t-k} = [X_{t-1}, \ldots, X_{t-k}].$$

The Transfer Entropy (TE) from $\{X_t\}$ to $\{Y_t\}$ is

$$\mathrm{TE}_{X \to Y}(k, \ell) = I(Y_t; \mathbf{X}_{t-k} \mid \mathbf{Y}_{t-\ell}) \,.$$

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## Remark

- TE measures how much $Y_t$ depends on the past of $\{X_t\}$ once its past is already known.
- If $Y_t \perp \mathbf{X}_{t-k} \mid \mathbf{Y}_{t-\ell}$, then $\mathrm{TE}_{X \to Y}(k, \ell) = 0$
- $I(Y_t; \mathbf{X}_{t-k} \mid \mathbf{Y}_{t-\ell}) \neq I(X_t; \mathbf{Y}_{t-\ell} \mid \mathbf{X}_{t-k}) \Rightarrow$ directional

# Diffusion process

- For r.v. $X \sim p_X$, construct a diffusion process $\{X_t\}_{t\in[0,T]}$ s.t.

$$X_0 \sim p_X, \qquad X_T \sim p_{X_T},$$

  with $p_{X_T}$ chosen to be easy to sample from.
- The process $\{X_t\}_{t\in[0,T]}$ follows

$$\begin{cases} dX_t = f_t X_t \, dt + g_t \, dW_t, \\ X_0 \sim p_X, \end{cases}$$

  where $f_t \leq 0$ and $g_t \geq 0$.
- Each $X_t$ has density $p_{X_t}$ and score $S^{p_{X_t}}(x) := \nabla_x \log p_{X_t}(x)$.

# KL Representation via Scores

- If another density $q_X$ is evolved under the same diffusion dynamics as $p_X$, then by decomposition

$$D_{\mathrm{KL}}[p_X \,\|\, q_X] = \int_0^T \frac{g_t^2}{2} \, \mathbb{E}_{x \sim p_{X_t}} \left[ \| S^{p_{X_t}}(x) - S^{q_{X_t}}(x) \|^2 \right] dt + D_{\mathrm{KL}}[p_{X_T} \,\|\, q_{X_T}].$$

where $\| \cdot \|$: Euclidean norm in $\mathbb{R}^{N_X}$.

This links KL divergence to the mismatch between the time-dependent score functions along the diffusion path.

# KL Estimation with Learned Scores

- In practice, true scores are approximated by neural networks.

  For a generic density $p$, the score network is trained by denoising score matching:

$$\theta^\star = \arg\min_\theta \int_0^T \mathbb{E}_{x\sim p}\mathbb{E}_{\tilde{x}\sim p_{0t}(\cdot|x)}\left[\left\|S^{p_{X_t}}(\tilde{x};\theta) - S^{p_{0t}}(\tilde{x}\mid x)\right\|^2\right] dt.$$

- Using learned scores, the KL divergence is estimated by

$$e(p,q) = \int_0^T \frac{g_t^2}{2}\,\mathbb{E}_{x\sim p_{X_t}}\left[\left\|S^{p_{X_t}}(x;\theta_1^\star) - S^{q_{X_t}}(x;\theta_2^\star)\right\|^2\right] dt.$$

  This is the first term of the KL decomposition, replacing true scores with learned scores, and typically

$$e(p,q) \approx D_{\mathrm{KL}}[p \,\|\, q].$$

# Entropy Estimation with Learned Scores

**Definition**

Entropy of r.v. $X$: $H(X) = \mathbb{E}_{x \sim p_X}\big[-\log p_X(x)\big]$

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## Remark

Gaussian case: if $X \sim \varphi_\sigma \triangleq \mathcal{N}(0, \sigma^2 I_{N_x})$, then

$$H(X) = \frac{N_x}{2}\log(2\pi\sigma^2) + \mathbb{E}_{x\sim p_X}\left[\frac{\|x\|^2}{2\sigma^2}\right] - D_{\mathrm{KL}}(p_X \,\|\, \varphi_\sigma)\,.$$

$\Rightarrow$ Score-based entropy estimator:

$$H(X;\sigma) \simeq \frac{N_x}{2}\log(2\pi\sigma^2) + \mathbb{E}_{x\sim p_X}\left[\frac{\|x\|^2}{2\sigma^2}\right] - e(p_X, \varphi_\sigma) - \frac{N_x}{2}\left(\log(\chi_T) - 1 + \frac{1}{\chi_T}\right).$$

where $\chi_t = \left(k_t^2\sigma^2 + k_t^2\int_0^t \frac{g_s^2}{k_s^2}\,ds\right), \quad k_t = \exp\left(\int_0^t f_s\,ds\right), \quad t \in [0, T].$

# CMI Estimation with Score Discrepancy

Since CMI can be written in three ways:

$$\begin{aligned} I(X;Y \mid Z) &= \mathbb{E}_{(y,z)\sim p_{YZ}}\left[D_{\mathrm{KL}}\left(p_{X|y,z} \,\|\, p_{X|z}\right)\right] \\ &= H(X \mid Z) - H(X \mid Y,Z) \\ &= I(X;Y,Z) - I(X;Z), \end{aligned}$$

CMI can be estimated with score discrepancy $e(\cdot,\cdot)$:

$$\mathbb{E}_{(y,z)\sim p_{YZ}}\left[D_{\mathrm{KL}}\left(p_{X|y,z} \,\|\, p_{X|z}\right)\right] \approx \mathbb{E}_{\substack{(y,z)\sim p_{YZ}\\ x\sim p_{X|y,z}}}\left[e\left(p_{X|y,z},\, p_{X|z}\right)\right]$$

$$H(X \mid Z) - H(X \mid Y,Z) \approx \mathbb{E}_{\substack{(y,z)\sim p_{YZ}\\ x\sim p_{X|y,z}}}\left[e\left(p_{X|y,z},\, \varphi_\sigma\right)\right] - \mathbb{E}_{z\sim p_Z}\left[e\left(p_{X|z},\, \varphi_\sigma\right)\right]$$

$$I(X;Y,Z) - I(X;Z) \approx \mathbb{E}_{\substack{(y,z)\sim p_{YZ}\\ x\sim p_{X|y,z}}}\left[e\left(p_{X|y,z},\, p_X\right)\right] - \mathbb{E}_{z\sim p_Z}\left[e\left(p_{X|z},\, p_X\right)\right]$$

$$\approx \left[\mathbb{E}_{\substack{(y,z)\sim p_{YZ}\\ x\sim p_{X|y,z}}}\left[e\left(p_{X|y,z},\, p_X\right)\right] - e(p_X,\varphi_\sigma)\right] - \left[\mathbb{E}_{z\sim p_Z}\left[e\left(p_{X|z},\, p_X\right)\right] - e(p_X,\varphi_\sigma)\right]$$

# TE Estimation

## Dataset

Source series: $\{X_t \in \mathbb{R}^{N_x}\}_{t=1}^{T}$; target series: $\{Y_t \in \mathbb{R}^{N_y}\}_{t=1}^{T}$. Choose source and target lags $k, \ell \in \mathbb{N}$.
For time index $t > \max(k, \ell)$, define

$$Y := Y_t, \quad X := [X_{t-1}, \ldots, X_{t-k}] \in \mathbb{R}^{kN_x}, \quad Z := [Y_{t-1}, \ldots, Y_{t-\ell}] \in \mathbb{R}^{\ell N_y}.$$

Stacking them for $t = \max(k, \ell) + 1, \ldots, T$ gives the dataset

$$\left\{ \left( X^{(i)}, Y^{(i)}, Z^{(i)} \right) \right\}_{i=1}^{T-\max(k,\ell)}.$$

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## TE

- $\mathrm{TE}_{X \to Y}(k, \ell) = I(Y; X \mid Z)$.
- $\mathrm{TE}_{Y \to X}(\ell, k)$: by swapping $\{X_t\}$ and $\{Y_t\}$.

# Algorithm

- $c$: only use conditional scores

$S_{x,z}^{p_{Y_{t^\star}}} \leftarrow net_\theta([Y_{t^\star}, X, Z], t^\star, [1,0,0])$
$S_{z}^{p_{Y_{t^\star}}} \leftarrow net_\theta([Y_{t^\star}, X, Z], t^\star, [1,-1,0])$
**if** $option = 1$ **then**
  $\hat{I} \leftarrow T\frac{g_{t^\star}^2}{2}\left\|S_{x,z}^{p_{Y_{t^\star}}} - S_{z}^{p_{Y_{t^\star}}}\right\|^2$ Eq. (15)
**else**
  $\chi_{t^\star} \leftarrow \left(k_{t^\star}^2\sigma^2 + k_{t^\star}^2\int_0^{t^\star} k_s^{-2}g_s^2\,\mathrm{d}s\right)$
  $I_1 \leftarrow \left\|S_{x,z}^{p_{Y_{t^\star}}} + \frac{Y_{t^\star}}{\chi_{t^\star}}\right\|^2$
  $I_2 \leftarrow \left\|S_{z}^{p_{Y_{t^\star}}} + \frac{Y_{t^\star}}{\chi_{t^\star}}\right\|^2$
  $\hat{I} \leftarrow T\frac{g_{t^\star}^2}{2}\left[I_1 - I_2\right]$ Eq. (16)

- $j$: jointly use both marginal and conditional scores

$S_{y,z}^{p_{X_{t^\star}}} \leftarrow net_\theta([Y_{t^\star}, X, Z], t^\star, [1,0,0])$
$S_{z}^{p_{X_{t^\star}}} \leftarrow net_\theta([Y_{t^\star}, X, Z], t^\star, [1,-1,0])$
$S^{p_{X_{t^\star}}} \leftarrow net_\theta([[Y_{t^\star}, X, Z], t^\star, [1,-1,-1])$
**if** $option = 1$ **then**
  $I_1 \leftarrow \left\|S_{x,z}^{p_{Y_{t^\star}}} - S^{p_{Y_{t^\star}}}\right\|^2$
  $I_2 \leftarrow \left\|S_{z}^{p_{Y_{t^\star}}} - S^{p_{Y_{t^\star}}}\right\|^2$
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**else**
  $\chi_{t^\star} \leftarrow \left(k_{t^\star}^2\sigma^2 + k_{t^\star}^2\int_0^{t^\star} k_s^{-2}g_s^2\,\mathrm{d}s\right)$
  $I_1 \leftarrow \left\|S_{x,z}^{p_{Y_{t^\star}}} + \frac{Y_{t^\star}}{\chi_{t^\star}}\right\|^2 - \left\|S^{p_{Y_{t^\star}}} + \frac{Y_{t^\star}}{\chi_{t^\star}}\right\|^2$
  $I_2 \leftarrow \left\|S_{z}^{p_{Y_{t^\star}}} + \frac{Y_{t^\star}}{\chi_{t^\star}}\right\|^2 - \left\|S^{p_{Y_{t^\star}}} + \frac{Y_{t^\star}}{\chi_{t^\star}}\right\|^2$
  $\hat{I} \leftarrow T\frac{g_{t^\star}^2}{2}\left(I_1 - I_2\right)$ Eq. (18)

# Synthetic datasets

## Linear Gaussian system

$$\begin{cases} x_t = b_x x_{t-1} + \lambda y_{t-1} + \varepsilon_t^x \\ y_t = b_y y_{t-1} + \varepsilon_t^y, \end{cases}$$

$\varepsilon_t^x \sim \mathcal{N}(0, \sigma_x^2),\ \varepsilon_t^y \sim \mathcal{N}(0, \sigma_y^2)$

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## Joint system

$$y_t = \begin{cases} z_{t-1}, y_{t-1} < \lambda \\ \rho x_{t-1} + \sqrt{1-\rho^2} z_{t-1}, y_{t-1} \geq \lambda \end{cases}$$

$x_t, z_t \sim N(0,1),\ \rho \in (-1,1),\ y_t$

# Sample size effect

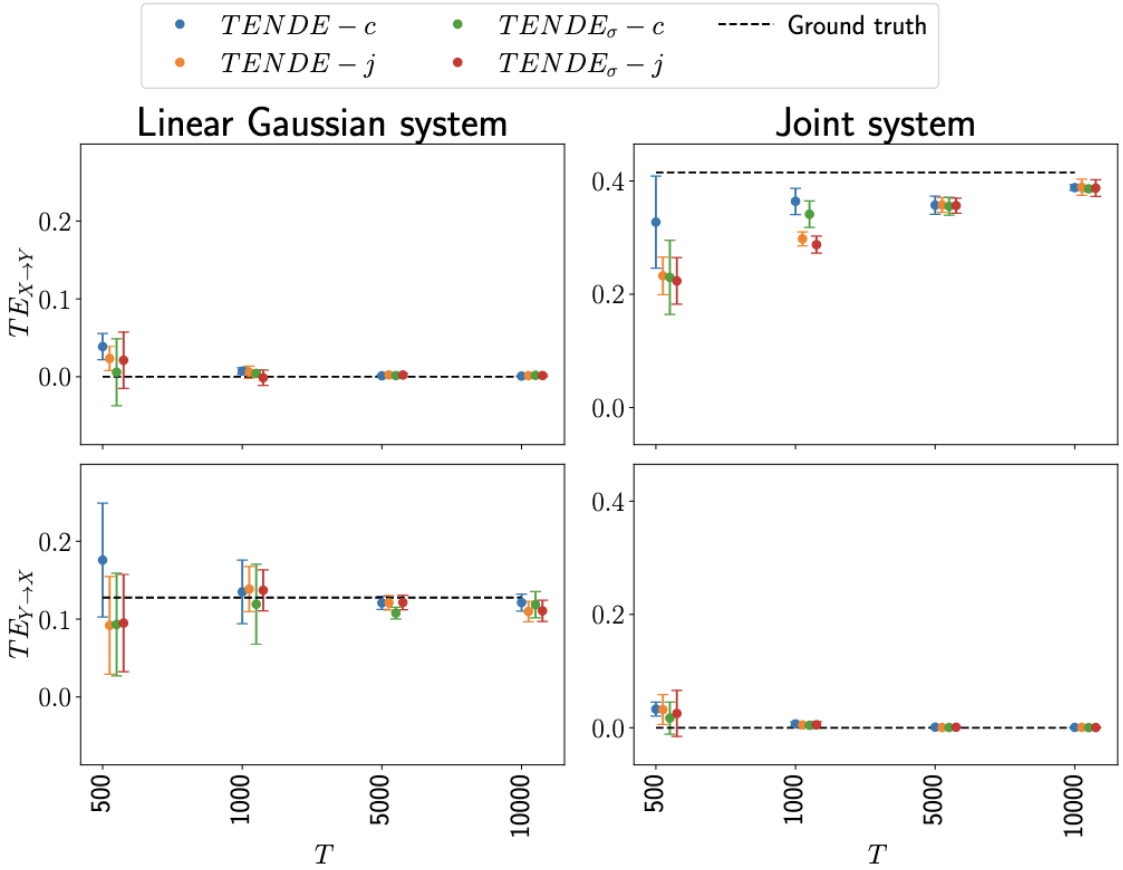


**Figure 1:** TE estimation across sample sizes.

# Consistency

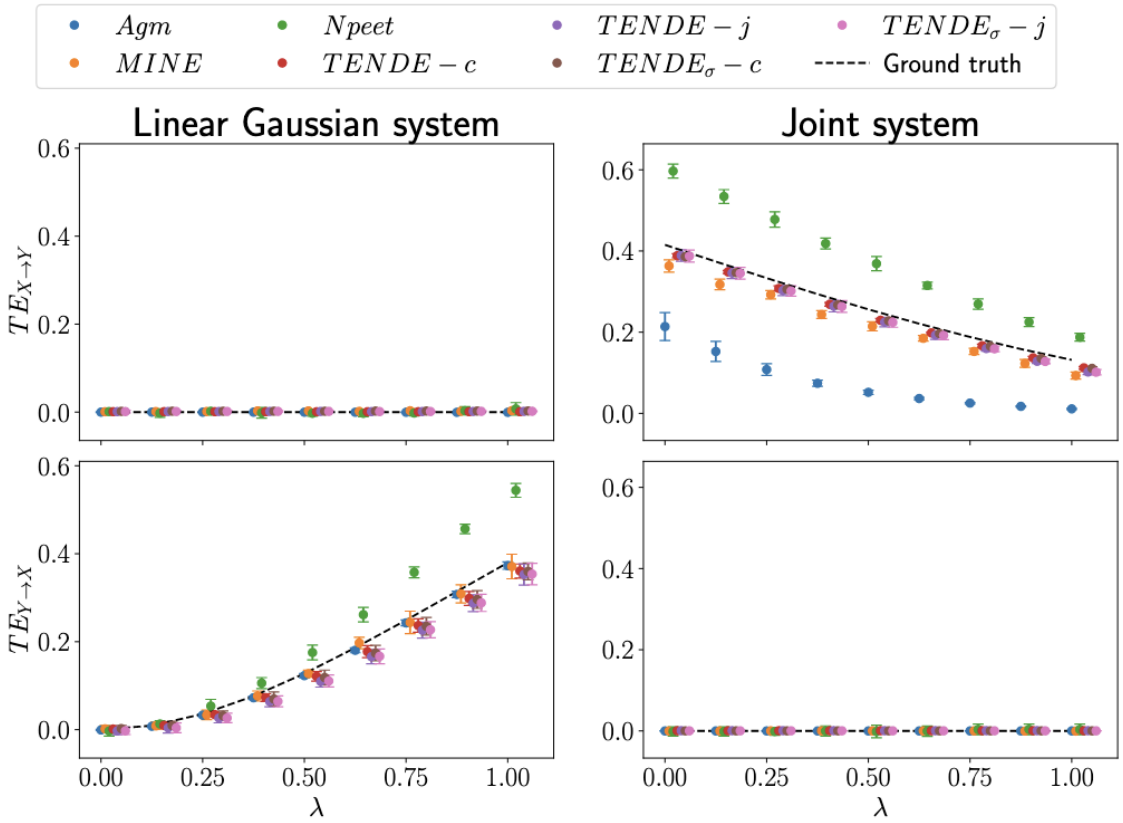


**Figure 2:** TE estimation for varying coupling strength.

# Redundant stacking

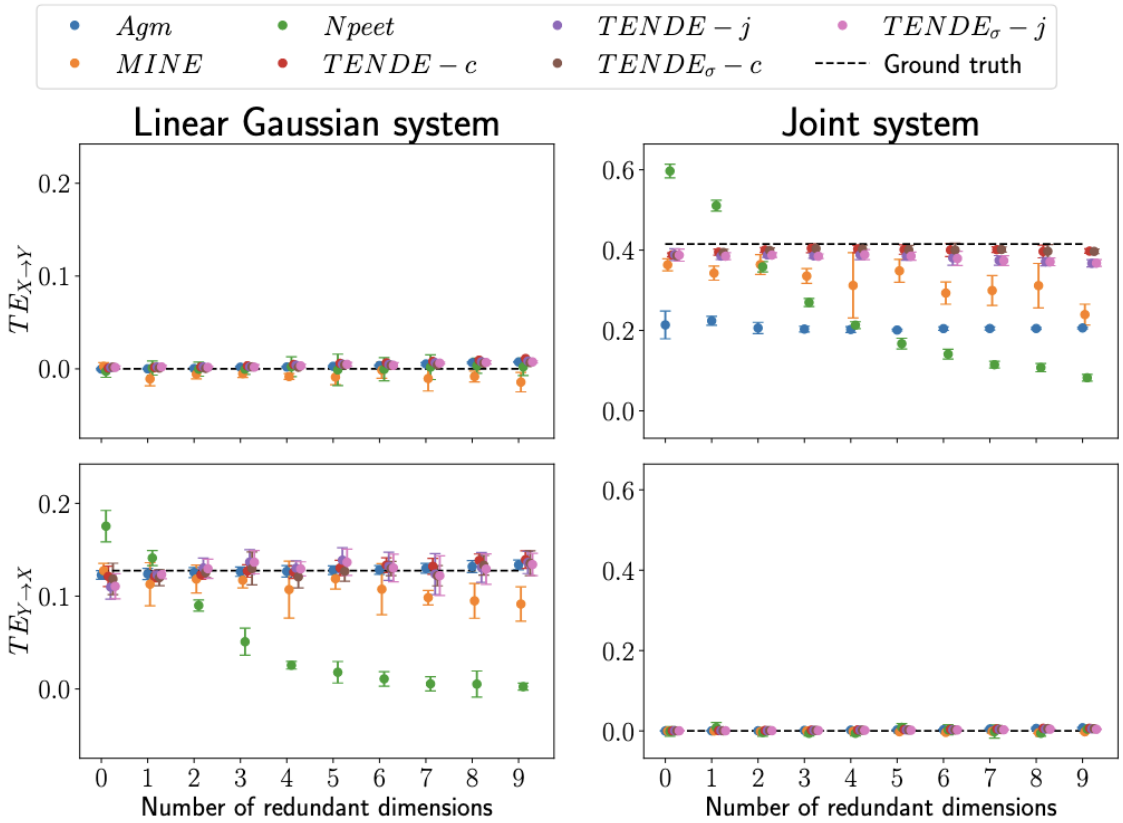


**Figure 3:** TE estimation with added redundant (noise) dimensions.

# Linear stacking

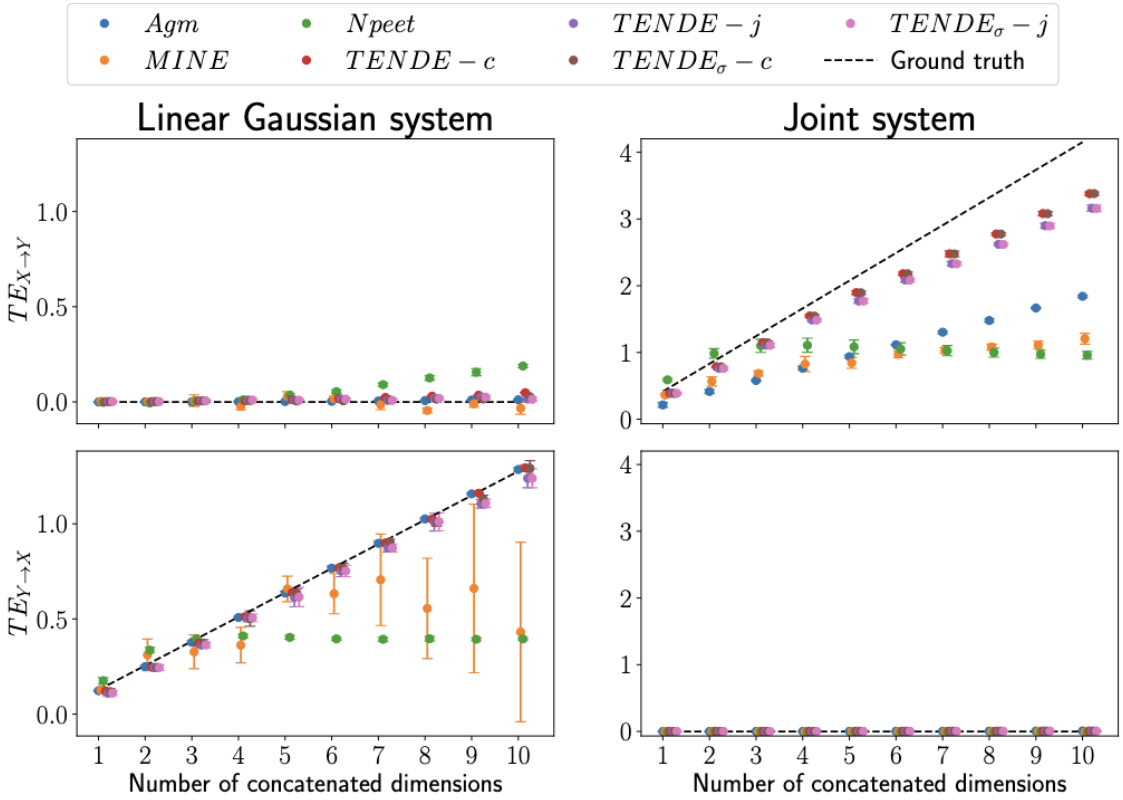


**Figure 4:** TE estimation for linearly stacked systems where multiple independent process copies create additive transfer entropy.

# Real dataset

## Santa Fe physiological dataset

Heartbeat and respiration force time series, sampled every 0.5 seconds.

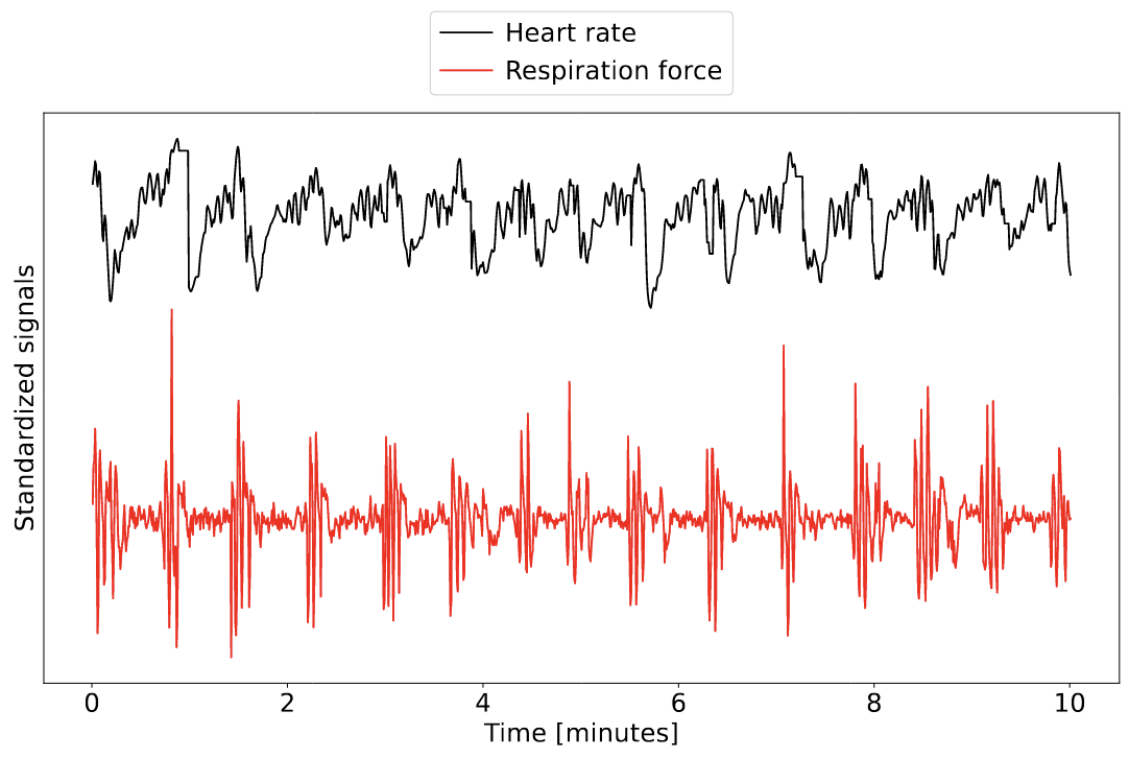


**Figure 5:** Sampled time series over 10 minutes.

# TE between breathing and heart signals

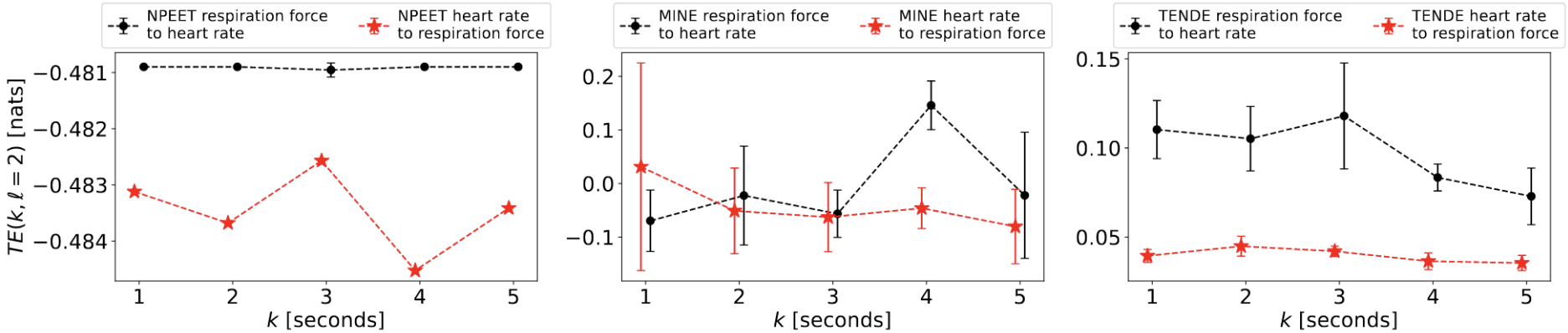


**Figure 6:** $TE(k, \ell = 2)$ as a function of lag $k$

- The TE from respiration force to heart rate is consistently higher than the reverse direction across the tested lags.

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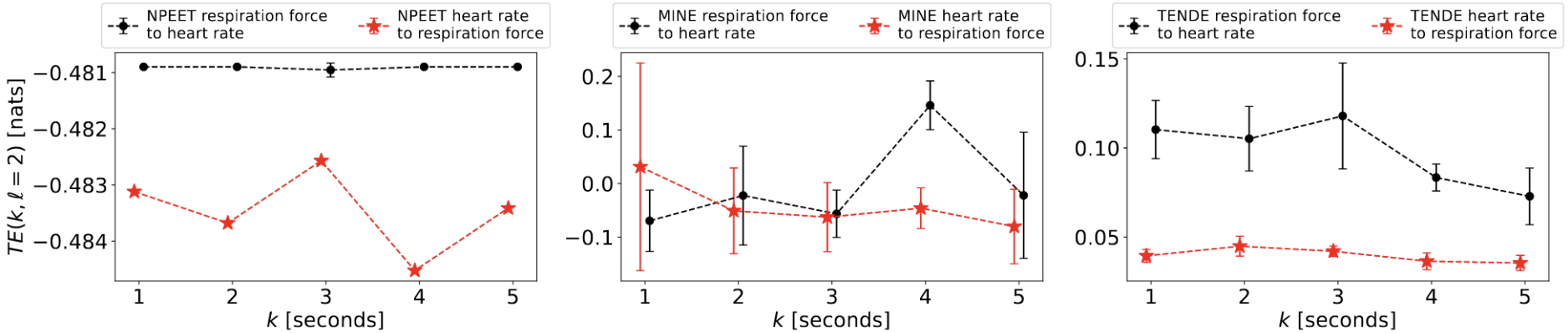


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- asymmetry: TE drops beyond 3s in the respiratory-to-heart direction, while the reverse direction remains relatively stable across lags.

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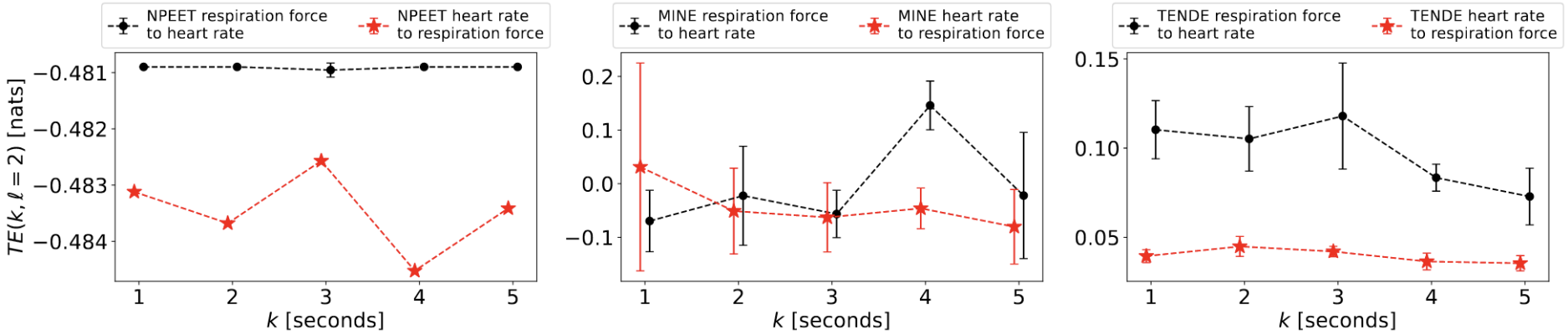


**Figure 6:** $TE(k, \ell = 2)$ as a function of lag $k$

- The TE from respiration force to heart rate is consistently higher than the reverse direction across the tested lags.
- asymmetry: TE drops beyond 3s in the respiratory-to-heart direction, while the reverse direction remains relatively stable across lags.
- TENDE's estimates are smoother and more plausible than competitors.

# Conclusion

1. TENDE: a novel approach that leverages score-based diffusion models for transfer entropy estimation via conditional mutual information with minimal assumptions.

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1. TENDE: a novel approach that leverages score-based diffusion models for transfer entropy estimation via conditional mutual information with minimal assumptions.
2. TENDE is accurate, robust, and interpretable on synthetic benchmarks and realworld datasets.